

failures. This gave rise to a need for a proactive approach to get notified of any anomalies in assets. With the introduction of IoT platforms, asset health data can be analysed in real time, and notifications can be sent ahead of time.

A life safety solutions company, called NAFFCO, based out of the UAE identified the need for proactive monitoring of solution equipment for

increased uptime and reliability. It collaborated with IoT solution providers for deploying an inspection system. The system transfers all data via a gateway to the IoT platform for real-time monitoring. Whenever there is any deviance, an alarm is triggered, which is converted in an e-mail or SMS and sent to concerned members. The solution has enabled NAFFCO to monitor its safety equipment 24/7.

Businesses of any scale require strategies to reduce operational expenses on their assets in the long run. This requires proactive and predictive maintenance. Figuring out asset worth, loss in time due to unplanned shutdown, expense of purchasing new asset and other associated business hassles, the ROI of automated inspection systems can be clearly deduced.

—Deeksha Sharma, technology journalist, EFY

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GREEN TECH:

“An EV Is About EIGHT TO TEN TIMES CHEAPER As Compared To An IC CAR”

➤ Electric vehicles (EVs) are expected to be more economical than internal combustion (IC) engine cars in the long run and, of course, more environment-friendly. But four-wheeler EVs have still not managed to hit the high note in India in terms of demand and customer adoption. What is the root cause of this situation? What options do Indian purchasers have? Paromik Chakraborty of *Electronics For You* spoke with Greg Moran, co-founder and chief executive officer, Zoomcar, to get these questions answered and to learn about Zoomcar's venture with EVs

Q. Are four-wheeler EVs affordable for Indian consumers right now?

A. EVs have been around in India for a while now. But these just could not stir up the Indian customers. One of the biggest reasons for this is something known as range anxiety. After spending ₹ 600,000 to ₹ 700,000 on a car, the consumer wants to get at least 250 kilometres without worrying about recharging. But that is not the case yet. However, if we talk only about the economics of driving a car, an EV is about eight to ten times cheaper as compared to an IC car.

Q. How are EVs more economical than traditional cars?

A. Here is the math. Currently, petrol price is hovering somewhere around the ₹ 80 per litre zone. A small city car—like an entry-level or premium hatchback like Maruti Swift or Hyundai i10—provides a fuel average of 16 to 20 kilometres per litre, which brings the running cost to around

₹ 5 per kilometre for a petrol car, and around ₹ 4 per kilometre for a diesel car.

In addition to this, there are ser-

vice and maintenance costs around of ₹ 0.60 per kilometre for petrol cars and about ₹ 1 per kilometre for diesel cars. Hence, on an average, cost of driving an IC car is around ₹ 6 per kilometre.

On the other hand, for Mahindra Electric P6 variant the cost comes out to be ₹ 0.80 per kilometre when we take the cost of electricity at an estimate of ₹ 6 per unit and the range of 10 kilometres per kilowatt-hour. This is about eight times cheaper than for an IC car. Hence, operating an EV is definitely affordable. But due to range anxiety and better and superior alternatives in IC engines for the same price, it is getting difficult for EVs to make it into the lives of Indian consumers.

Q. How can this range anxiety be addressed?

A. Subscription is the way ahead for EVs to address range anxiety. Consumers can get the taste of first blood



Greg Moran, co-founder and chief executive officer, Zoomcar